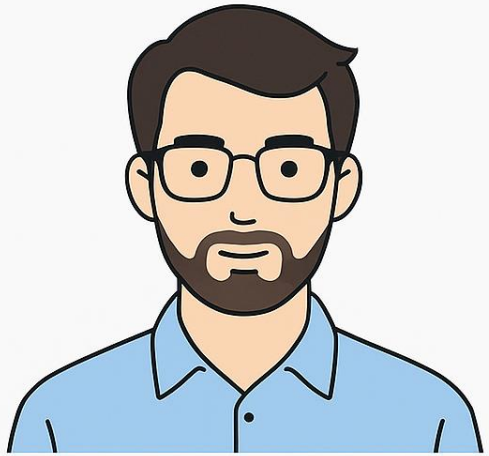
Autonomous Real-Time Threat and Anomaly Detection for IoT Security in 5G-Enabled Urban Surveillance Systems

PROFILE



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INTRODUCTION

The rapid emergence of 5G-enabled smart cities has driven the widespread deployment of IoT devices such as cameras, sensors, and controllers, enhancing urban functionality but also exposing these systems to a wide range of cyber threats including malware, denial-of-service attacks, data breaches, and zero-day exploits. Traditional, centralized security approaches often lack the real-time responsiveness and adaptability needed to detect novel or hidden anomalies, making them insufficient in dynamic urban environments. To address this, there is an urgent need for autonomous, real-time, and privacy-preserving cybersecurity solutions. This project introduces an AI-based intrusion detection system designed to operate intelligently and independently within smart city IoT networks, providing proactive protection against evolving threats.

EMPATHY

In today's deeply interconnected urban world, a hacked surveillance camera that misses a crime, a tampered traffic signal that causes a devastating accident, or a compromised sensor that fails to detect danger—these are not just system errors; they are moments where human lives are put at risk. Behind every glitch lies a person—a child crossing the street, a family waiting at a signal, a community trusting the unseen systems around them. When these systems fail, it's not just data that is lost, but a sense of safety, dignity, and trust. As our cities become smarter, the threats they face become more silent, more invisible—but their impact is painfully real. This project isn't just about upgrading technology; it's about standing guard for the people in our cities, protecting their everyday moments with intelligent systems that never sleep, never look away, and never stop caring.

BACKGROUND & MOTIVATION

With the rise of 5G-enabled smart cities, thousands of IoT devices like surveillance cameras and traffic sensors are being deployed to improve urban infrastructure. However, these systems are often vulnerable to cyber threats, and traditional centralized security approaches fail to provide real-time, scalable protection. The growing complexity of attacks demands smarter, decentralized solutions that can adapt quickly while maintaining data privacy.

The project is motivated by the need to secure smart city IoT systems through intelligent, real-time threat detection. By using deep learning and federated learning, the aim is to build a system that can detect and respond to attacks autonomously without relying on human intervention or centralized control. This ensures better safety, privacy, and resilience in critical urban networks.

LITERATURE REVIEW

Paper Title	Methodology/ Implementation	Data set Description	Performance Parameter	Limitations/ Disadvtanges	Gaps Identified
1. AI-Driven Anomaly Detection for Securing IoT Devices in 5G-Enabled Smart Cities	Hybrid Deep Learning (AE + LSTM + CNN) with Federated Learning and SHAP Explainability	TON_IoT, CICIDS2017, UNSW-NB15, + synthetic adversarial attacks (FGSM, PGD)	Precision: 97.5%, Recall: 96.2%, Inference <310 ms	Not tested in edge deployment; limited smart city simulation	Lacks practical edge simulation across 5G environments
2."IoT Anomaly Detection Methods and Applications: A Survey"	Survey of 64 research papers on ML/DL techniques	Not applicable (review paper)	Not applicable	No implementation or testing	No real-time testing, no FL, no smart city deployment

LITERATURE REVIEW

3.Attack and Anomaly Detection in IoT Sensors	Classical ML models: RF, DT, ANN, SVM, LR	Custom-collected IoT sensor data (multi-class attack labels)	Accuracy ~99.4% (RF best)	No DL or time-based models; poor generalization	No FL, no hybrid models, no real-time deployment
4.Anomaly Detection Using Deep Neural Network for IoT Architecture	Feedforward Deep Neural Network (DNN)	Smart home/synthetic IoT traffic	Accuracy ~97.5%	No LSTM, no sequence modeling or CNNs	No hybrid DL, no FL, no mitigation system

LITERATURE REVIEW

5. Adaptable Autoencoder for DDoS Mitigation	Autoencoder with adaptive thresholding + feature selection	CIC-DDoS2019 (modified)	Accuracy ~99.2%	Only for DDoS; no FL or privacy mechanism	Narrow focus; lacks general anomaly handling and edge testing
6.IDS on IoT with 5G Using Deep Learning	Unsupervised Autoencoder-based IDS	Custom 5G traffic from IoT networks	Model performance evaluated by accuracy ~99.76%	No comparison with hybrid DL or FL	No mitigation, no hybrid DL or FL

LITERATURE REVIEW

7.Poisoning Attacks on FL-based IDS	Federated Learning-based IDS with backdoor poisoning	Real IoT traffic from 46 edge devices	FL model was successfully poisoned	No defense proposed	No hybrid detection, no secure FL framework
8.FL for IoT Intrusion Detection	FL using ANN with multiple optimizers (FedAvg, FedAdam)	TON_IoT, CICIDS2017	Performance is evaluated using Accuracy ~96% (FedAvg best)	Only shallow ANN; lacks DL sequence learning	No hybrid models, no edge testing or mitigation

LITERATURE REVIEW

9.FL for Malware Detection in IoT Devices	FL with Autoencoder and MLP classifiers	Dataset includes N-BaIoT (9 malware-infected IoT devices)	Model performance is measured using accuracy ~97%, MLP ~95%	No mitigation system; only detection	Lacks explainability, no LSTM/CNN/real-time logic
10.AI Cybersecurity for Smart Cities (MDATA)	Knowledge graph-based model (MDATA) with alarm correlation	Simulated smart city cyber logs (no public dataset)	High detection rate; reduced false alarms	Complex, no DL/FL or real-time edge use	No DL/FL, lacks live smart city integration

LITERATURE REVIEW

11.Real-Time Anomaly Detection in Cold Chain IoT	LTE-M sensors → alert via SMS/email	Real deployment in vans	Reduced false alarms, efficient	No Deployment on Edge Devices	No Use of Federated Learning
12.Artificial Intelligence for Securing IoT Services in Edge Computing: A Survey	Survey of AI/ML techniques for IoT security in edge computing	Not applicable (survey)	Not applicable	No new experiments, heterogeneous works	Need standardized testbeds, lightweight models

LITERATURE REVIEW

13.Privacy-Preserving Video Anomaly Detection: A Survey	Survey and taxonomy of privacy-preserving VAD methods	Not applicable (survey)	Not applicable	Lack of standard privacy metrics, limited datasets	Need standard benchmarks, privacy metrics, on-device deployment
14.Privacy-Preserving Analysis for Remote Video Anomaly Detection in Real Life Environments	DP-based autoencoder compression + supervised anomaly detection	UCF-Crime	Accuracy/Precision/Recall tradeoffs by DP setting	Utility drop with stronger privacy, single dataset	More datasets, lightweight encoders, real-time latency studies

LITERATURE REVIEW

15. Machine and Deep Learning for IoT Security and Privacy: Applications, Challenges, and Future Directions	Survey of ML/DL methods for IoT security and privacy	Not applicable (survey)	Not applicable	Heterogeneous setups, resource cost issues	Lightweight privacy-preserving ML, standardized evaluation
16. Research on the Application of Artificial Intelligence Technology in the Field of Urban Security	Conceptual overview of AI use in urban security	Not applicable	Not applicable	High-level, lacks implementation details	Needs empirical validation, privacy-aware design

LITERATURE REVIEW

17.Neuro-Symbolic AI for IoT-Driven Smart Cities: A Next-Generation Framework for Urban Intelligence	Proposed neuro-symbolic AI framework for smart cities	Not applicable	Not applicable	Conceptual only, integration complexity untested	Prototype systems, benchmarks vs purely neural
18.AI-empowered IoT Security for Smart Cities	Survey + simulation of AI methods for smart-city IoT security	Synthetic/simulated data	Throughput, fairness, detection metrics from literature	Limited real-world deployment evidence	Need cross-layer, real deployments

LITERATURE REVIEW

19.Towards Machine Learning Based Intrusion Detection in IoT Networks	Pipeline comparing shallow ML vs DL for IoT IDS	NSL-KDD, DS2OS, IoTDevNet, IoTID20, IoT Botnet	Better Quantitative metrics	High compute cost for DL, limited generalization	Lightweight DL, real datasets, adversarial robustness
20.Impact of 5G Technology on Cybersecurity: A Comprehensive Systematic and Bibliometric Review	Systematic + bibliometric review of 5G security research	68 papers from major databases	Not applicable	Trends only, no experimental validation	Empirical studies on 5G attack surfaces, benchmarks

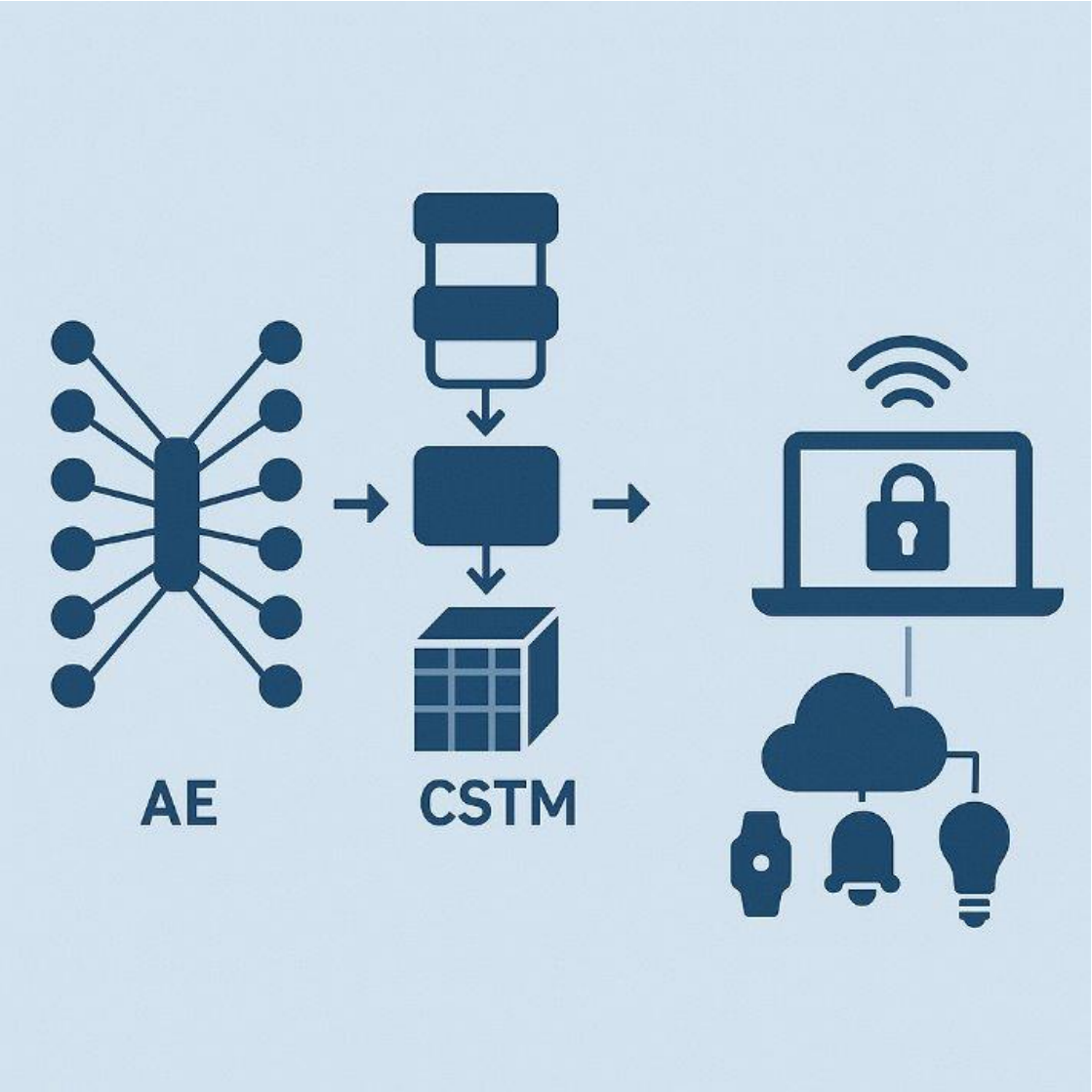
PROBLEM STATEMENT

"Autonomous Real-Time Threat and Anomaly Detection for IoT Security in 5G-Enabled Urban Surveillance Systems"

The rise of smart cities powered by 5G technology has led to a massive increase in IoT devices used for surveillance, traffic control, and public infrastructure. However, these devices are vulnerable to various cyber threats such as unauthorized access, data tampering, and zero-day attacks. Traditional security solutions are centralized and reactive, often failing to detect sophisticated threats in real time.

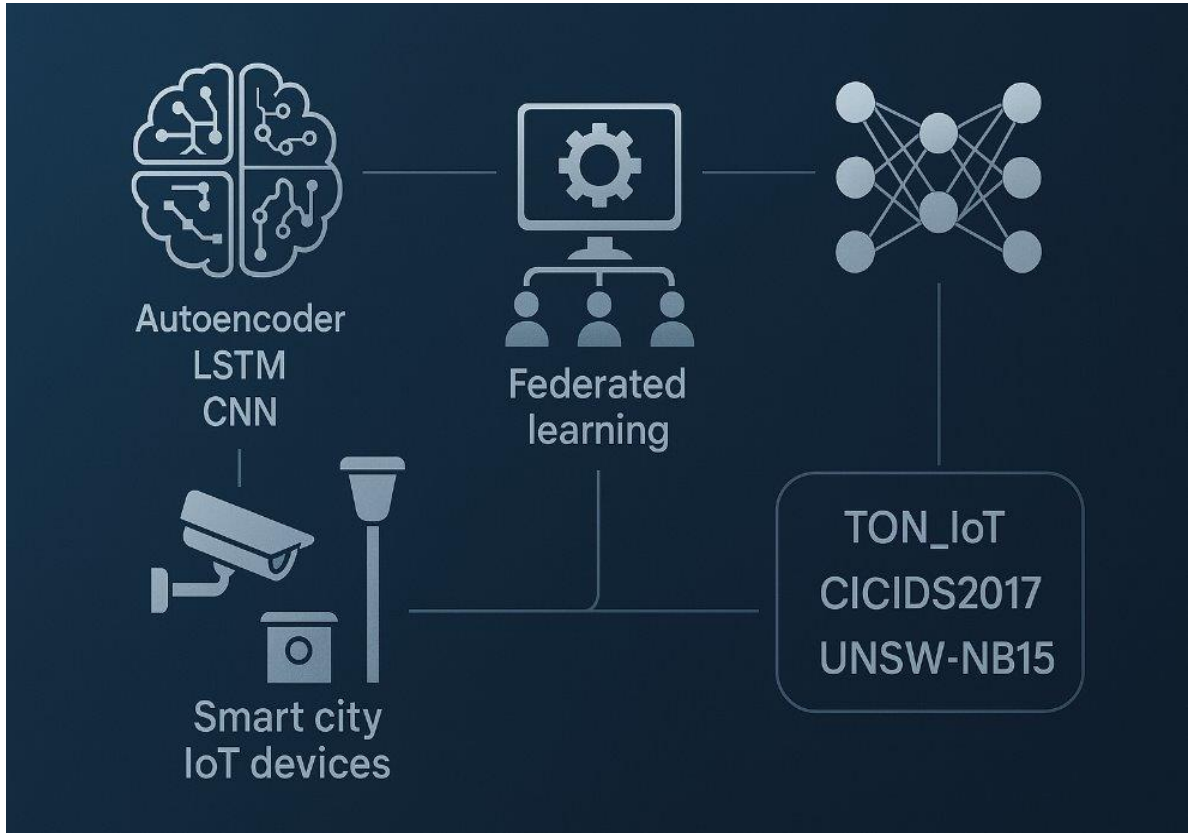
PROPOSED SOLUTION

This project proposes a hybrid deep learning-based threat detection system that combines Autoencoder, LSTM, and CNN for robust anomaly detection in IoT networks. Unlike existing centralized solutions, the model will be trained using federated learning to preserve data privacy .



OBJECTIVES

- **Develop a hybrid deep learning model** combining Autoencoder, LSTM, and CNN for detecting anomalies in IoT network traffic.
- **Implement federated learning** to enable decentralized model training while ensuring privacy of data across multiple nodes.
- **Use benchmark IoT intrusion datasets** (CICIDS2017) to train, validate, and test the model.
- **Evaluate the system's performance** in terms of accuracy, recall, and ROC-AUC, and optimize it for real-time detection.



TOOL LEARNING

- **Programming Language & Environment**

  Python (Jupyter Notebook)

- **Core ML Libraries**

  TensorFlow / Keras – Deep learning (Autoencoder, LSTM, CNN)

  scikit-learn – Preprocessing, baselines, metrics

- **Federated Learning Framework**

  TensorFlow Federated – Simulated federated training

- **Data Processing**

  pandas – Data cleaning & manipulation

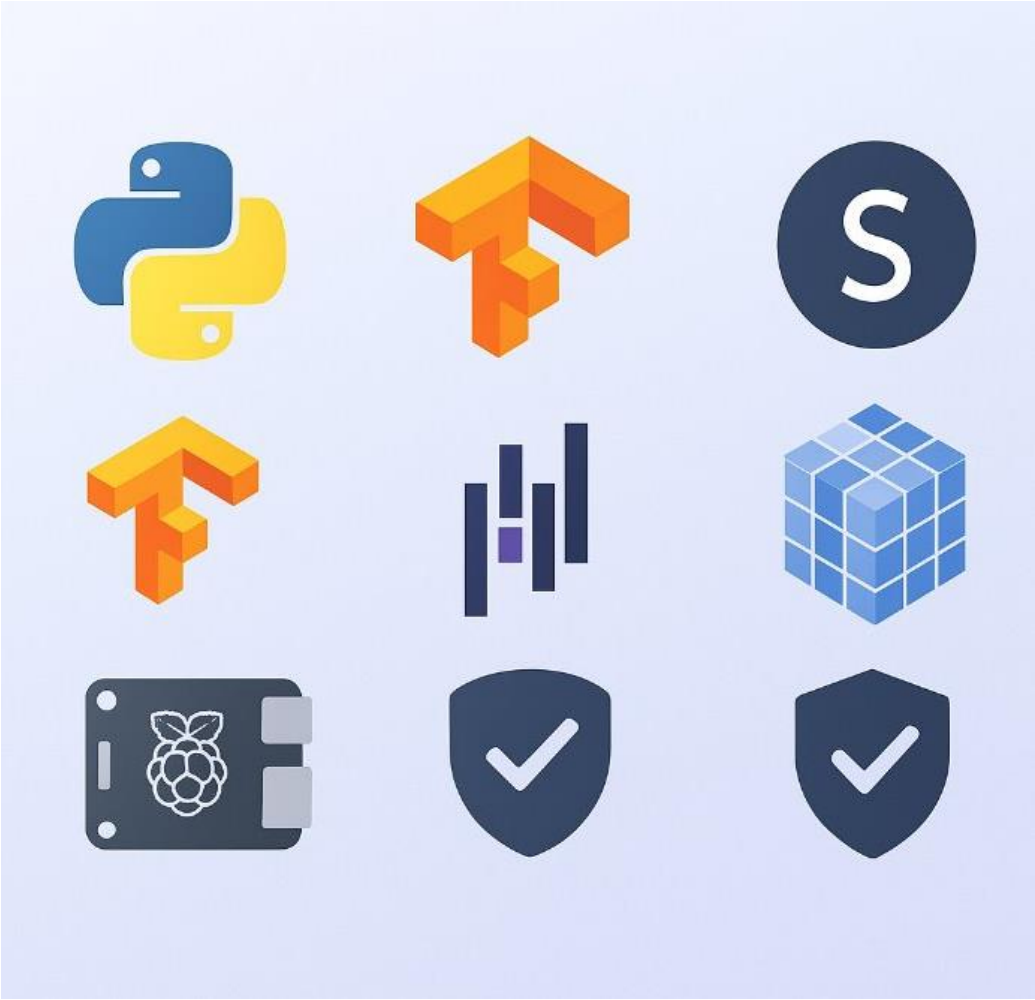
  NumPy – Numerical computations & vectorized operations

- **Deployment Target**

  Raspberry Pi – Edge inference

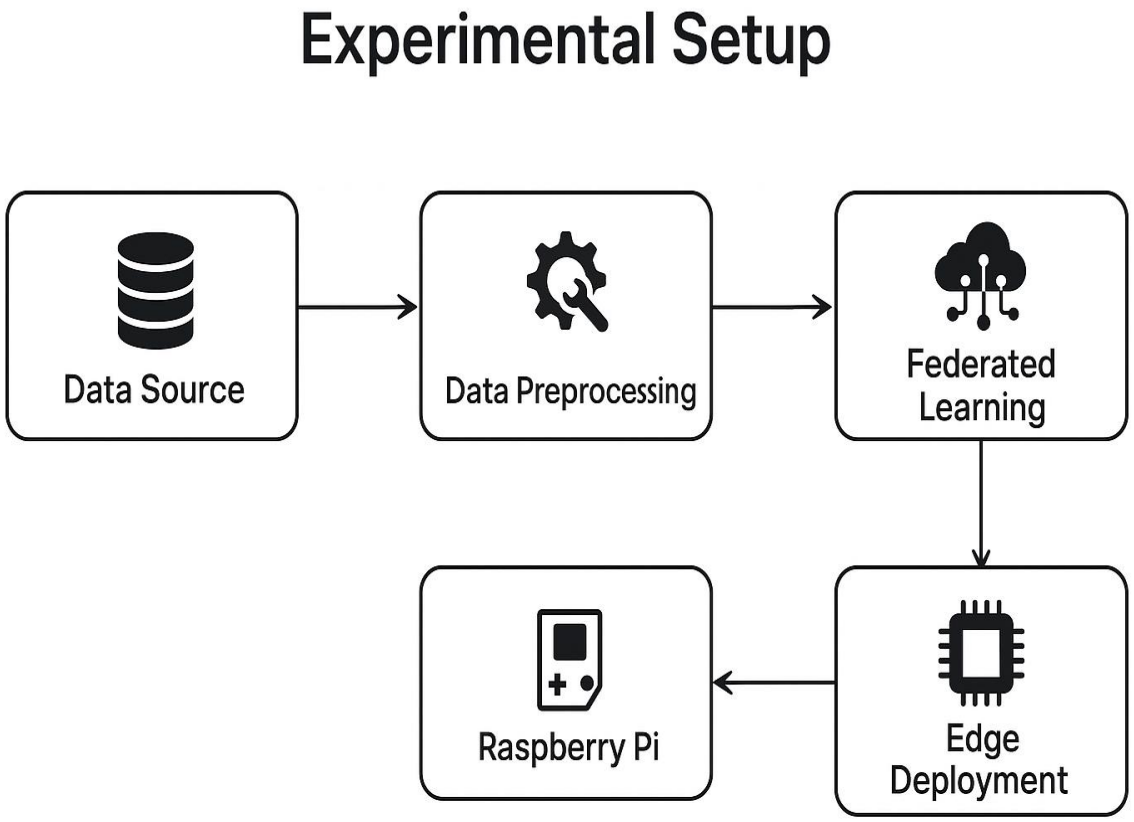
- **Privacy & Security**

  Federated Learning – Data stays local, only model updates shared



EXPERIMENTAL SETUP

- **Data Source** – You start with a public IoT/IDS dataset (like CICIDS2017) that contains network traffic data.
- **Data Preprocessing** – The raw data is cleaned, features are selected, and values are scaled so the model can understand it.
- **Federated Learning** – Instead of sending all data to one place, multiple simulated clients train the model locally and only share model updates, keeping the data private.
- **Edge Deployment** – The trained model is optimized and prepared for use on small, low-power devices.
- **Raspberry Pi** – The optimized model is deployed on a Raspberry Pi to run inference and detect anomalies directly at the edge.

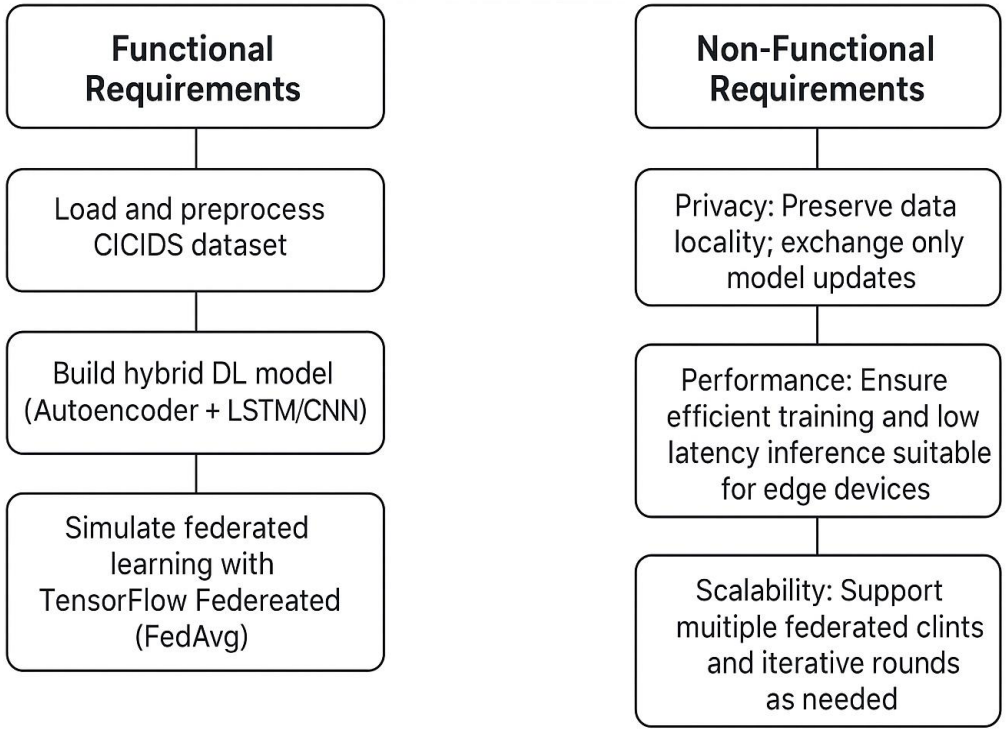


REQIUREMENT ANALYSIS

In requirement analysis, we describe what the system should do and the conditions it should meet.
It is divided into two main categories:

1. Functional Requirements: These describe the specific tasks and operations the system must perform.

2. Non-Functional Requirements: These define quality attributes of the system



ROADMAP

Problem Identification
Define the research problem: Autonomous real-time IoT threat & anomaly detection with Federated Learning

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Literature Survey
Study related works (2022–2025) on AI+ CNN + LSTM models, IoT security, and FL applica-

↓

Dataset Preparation
Select dataset (CICIDS2017, TON_IoT, etc.), load, clean, encode, and scale
Split into Train / Validation / Test sets

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Model Development
Build a hybrid AE + CNN + LSTM model for anomaly detection

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Federated Learning Setup
Implement federated learning for privacy-ping fraining

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Deployment
Convert model to TensorFlow Lite and deploy on Raspberry Pi for edge inference

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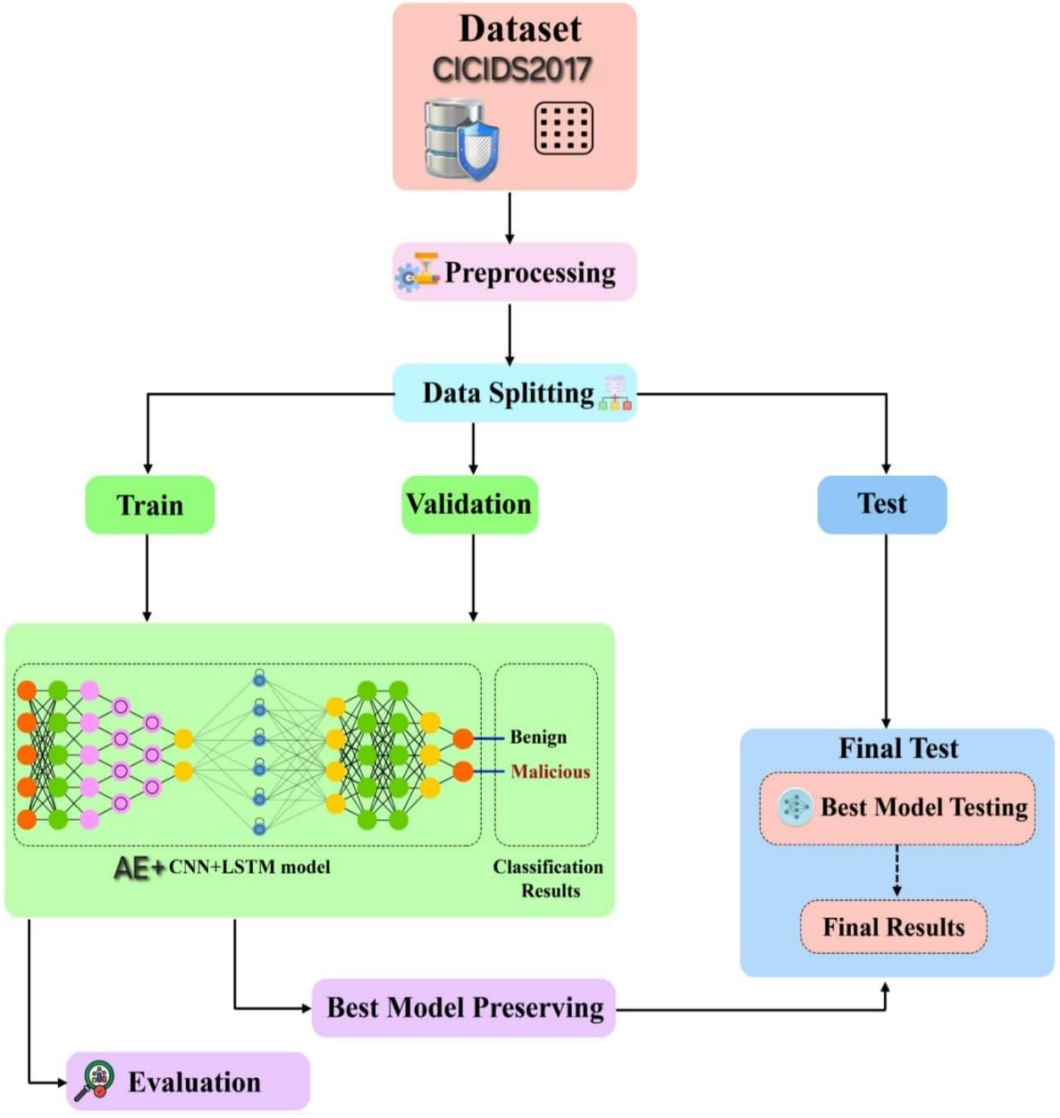
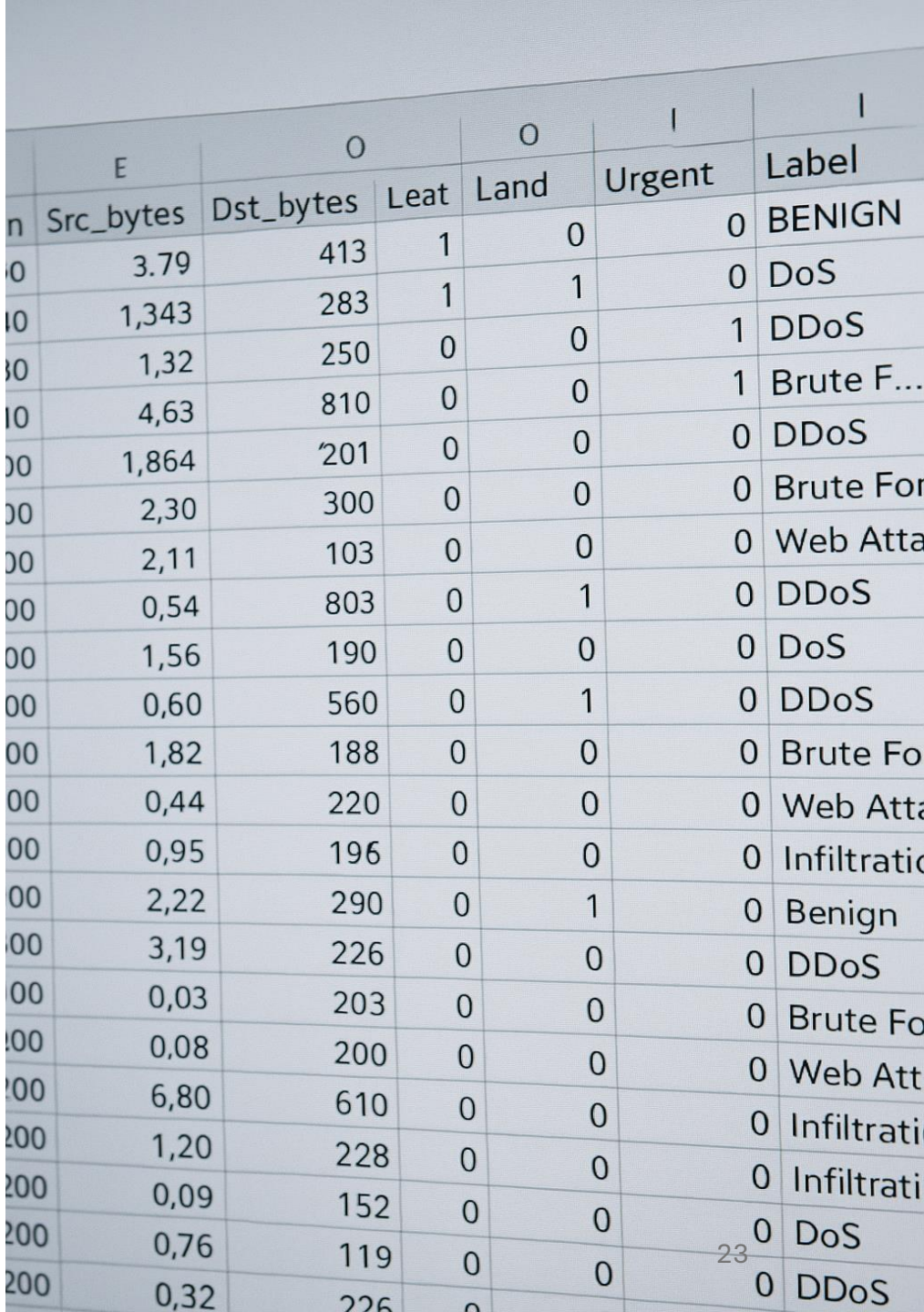
Documentation & Paper Writing
Document experiments, prepare research paper, and submit to journals/conferences



DATASET DESCRIPTION

- The CICIDS2017 dataset is a benchmark dataset for intrusion detection, created by the Canadian Institute for Cybersecurity. It contains 8 CSV files, each representing traffic captured on different days and attack scenarios. Across all files, the dataset has 80 features (network flow characteristics such as packet counts, durations, byte rates, and flag counts) plus a Label column indicating *BENIGN* or specific attack types.
- Example file statistics:
- Thursday-WorkingHours-Morning-WebAttacks.pcap_ISCX.csv – ~230,000 rows, 79 feature columns + Label
- Thursday-WorkingHours-Afternoon-Infiltration.pcap_ISCX.csv – ~170,000 rows, 78 feature columns + Label
- The dataset includes both benign traffic and multiple attack categories such as DoS, DDoS, Brute Force, Web Attacks, and Infiltration.



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n	Src_bytes	Dst_bytes	Leat	Land	Urgent	Label
0	3.79	413	1	0	0	BENIGN
0	1,343	283	1	1	0	DoS
0	1,32	250	0	0	1	DDoS
0	4,63	810	0	0	1	Brute F...
00	1,864	201	0	0	0	DDoS
00	2,30	300	0	0	0	Brute For
00	2,11	103	0	0	0	Web Atta
00	0,54	803	0	1	0	DDoS
00	1,56	190	0	0	0	DoS
00	0,60	560	0	1	0	DDoS
00	1,82	188	0	0	0	Brute Fo
00	0,44	220	0	0	0	Web Atta
00	0,95	196	0	0	0	Infiltratio
00	2,22	290	0	1	0	Benign
00	3,19	226	0	0	0	DDoS
00	0,03	203	0	0	0	Brute Fo
00	0,08	200	0	0	0	Web Att
00	6,80	610	0	0	0	Infiltrati
00	1,20	228	0	0	0	Infiltrati
00	0,09	152	0	0	0	DoS
00	0,76	119	0	0	0	DDoS
00	0,32	226	0			

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THANK YOU